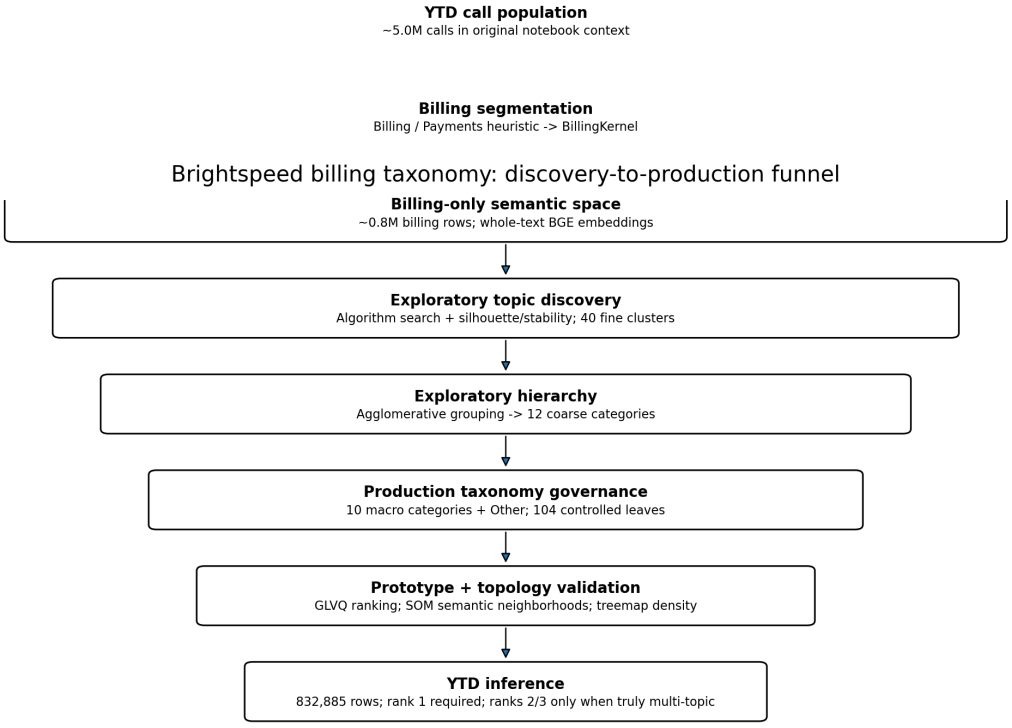
Brightspeed Billing Taxonomy — Process Report

> From 5 million raw calls to a governed production taxonomy of 10 macro categories, 104 subcategories, and 832,885 classified billing rows.

What this report covers

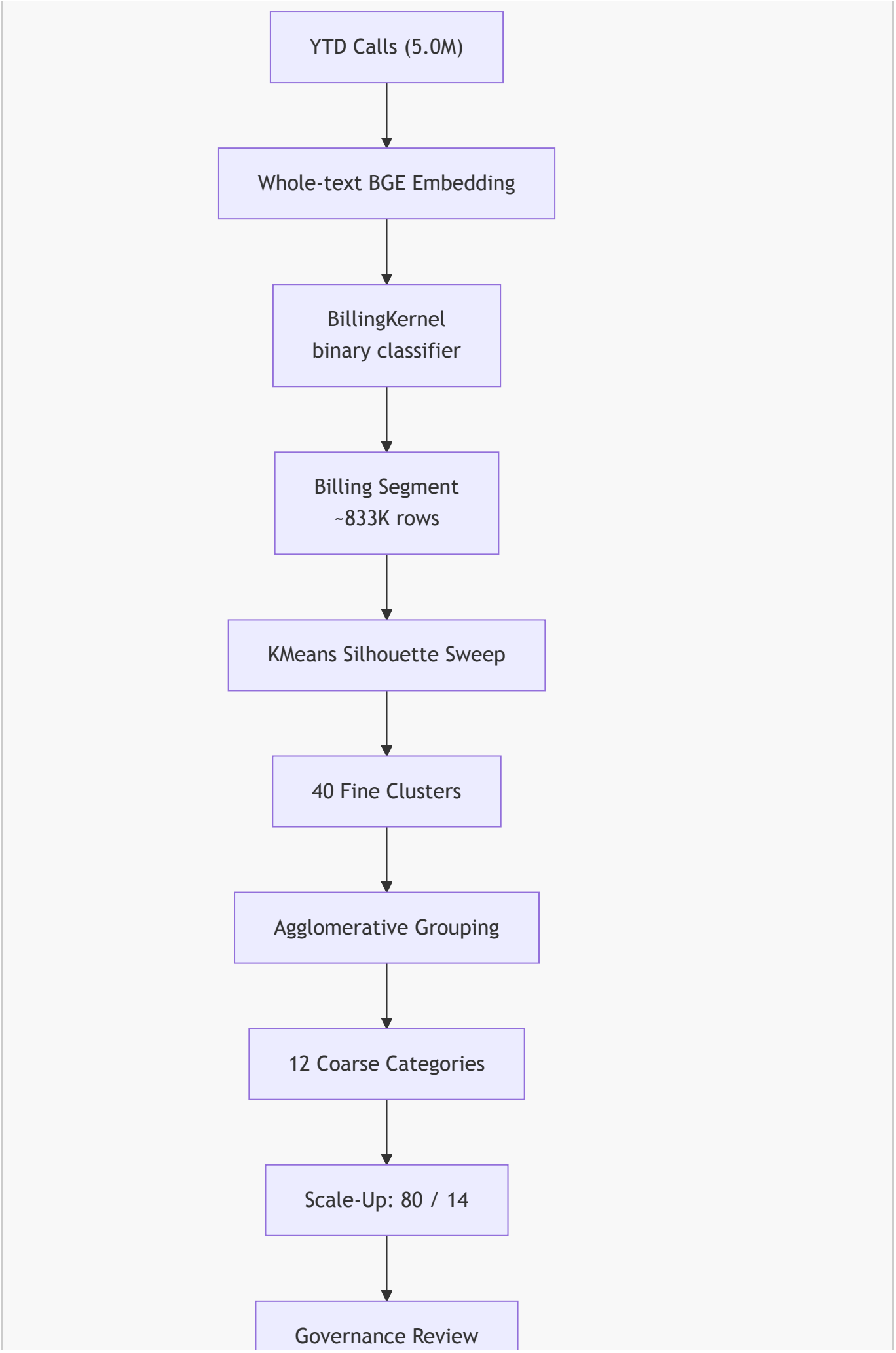
Brightspeed's billing taxonomy was built in stages — not handed down from a single clustering run. Each stage answers a different question, and the final output reflects business governance layered on top of data discovery.

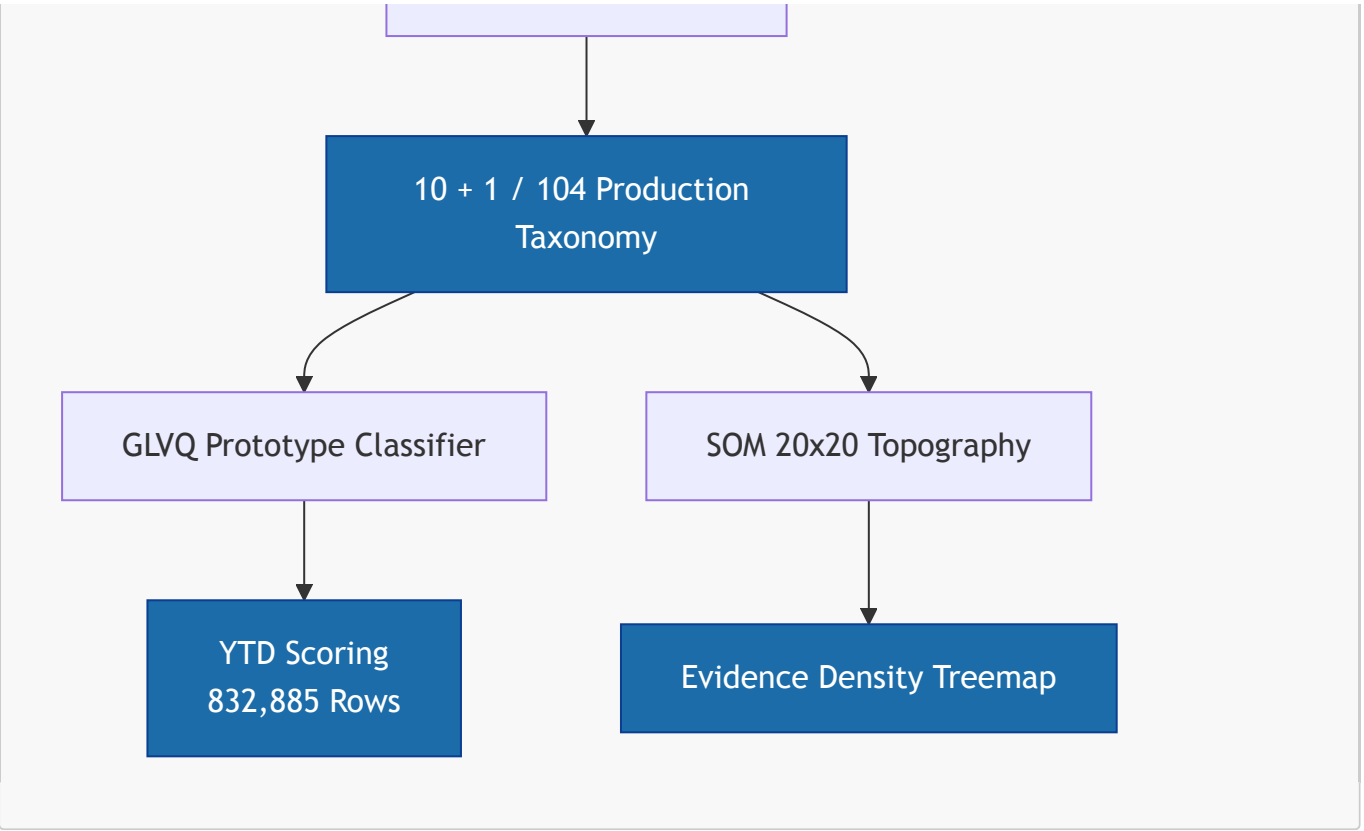
Stage	Structure	What it answers
Exploratory notebook	12 categories / 40 fine clusters	What does the billing topic landscape actually look like?
Scale-up engineering build	14 coarse groups / 80 fine clusters	What happens at higher discovery resolution?
Production taxonomy	**10 macro categories + Other / 104 canonical leaves**	**What stable vocabulary should classify all current and future billing contacts?**



12 / 40 was the data-discovery view. **10 + 1 / 104** is the governed production vocabulary. Silhouette and clustering help discover structure; they do not determine the final ontology alone.

How the pipeline flows


YTD Calls (5.0M)
Whole-text BGE Embedding
BillingKernel
binary classifier
Billing Segment
~833K rows
KMeans Silhouette Sweep
40 Fine Clusters
Agglomerative Grouping
12 Coarse Categories
Scale-Up: 80 / 14
Governance Review



Input and output files

Source data

File	Location	Description
`billing_all_ytd.xlsx`	`input_output/`	Raw YTD billing call data — all rows before classification. Contains `call_intent`, `category`, `month`, and account fields. This is the **input** to the pipeline.

Pipeline outputs

File	Location	Description
`billing_all_ytd_tagged.xlsx`	`input_output/`	Fully tagged output — every row with `ai_tag_1` / `ai_tag_2` / `ai_tag_3`, category, and subcategory. This is the **final deliverable**.
`billing_taxonomy_workbook.xlsx`	`artifacts/`	Review workbook with the canonical vocabulary, inference prompt, and non-technical usage notes

File	Location	Description
brightspeed_excel_inference_output.xlsx	artifacts/	Worked inference example showing tag assignment on sample rows

Model and taxonomy artifacts

File	Location	Description
canonical_vocabulary.csv	artifacts/	Allowed tag vocabulary: tag, category, subcategory, definition
tree_hierarchy.md	artifacts/	Complete 104-leaf taxonomy tree with evidence provenance
source_reason_to_canonical_mapping.csv	artifacts/	Maps source reason texts to canonical tags (111,836 mapped rows)
canonical_leaf_evidence_counts.csv	artifacts/	Observed evidence count per leaf
taxonomy_keywords.csv	artifacts/	Per-leaf TF-IDF keyword sets
curated_centroids_10_each.csv	artifacts/	10 curated representative texts per leaf
billing_inference_prompt.txt	artifacts/	Compact classification prompt for Excel-based tagging

Visual assets

File	Location	Description
process_funnel.png	assets/	Discovery-to-production funnel diagram
taxonomy_evolution.png	assets/	Taxonomy stage evolution
treemap_subcategories.png	assets/	Subcategory call-volume treemap
treemap_subcategory_evidence_density.png	assets/	Evidence-density treemap overlay
evidence_by_category.png	assets/	Macro-category evidence rollup
som_evidence_density.png	assets/	SOM evidence-density topology
som_map.png	assets/	SOM 20×20 cell-level map

Notebook

File	Location	Description
`01_billing_taxonomy_mining.ipynb`	root	End-to-end explanatory notebook: embedding → kernel → clustering → mining → GLVQ → SOM → scoring

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1. Embedding — one short intent, one vector

Every call's processed customer reason is treated as a single `call_intent` string and embedded whole — no sentence splitting, no chunking.

Model: `BAAI/bge-large-en-v1.5` · **Dimension:** 1024 · **Normalization:** L2 (cosine ≈ dot product)

```
"payment made but not reflected on account"  →  1024-d unit vector
"charged after service was disconnected"      →  1024-d unit vector
"need more time to pay the bill"              →  1024-d unit vector
```

---

2. BillingKernel — narrowing the search space

The first learned kernel is a **binary billing classifier**. It prevents non-billing calls (service, repair, sales) from polluting the billing topic model.

Architecture:

- L2-regularized logistic regression on normalized BGE embeddings
- Isotonic calibration for probability scaling
- Decision threshold tuned on held-out 10% for max F1
- OneClassSVM stored as a secondary positive-class reference

The original notebook saw roughly **5 million** YTD calls. After kernel filtering, the working billing segment is **832,885 rows**.

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3. Clustering — how many topics does the data support?

The notebook does not start by declaring 104 billing reasons. It first asks:

> *At what granularity do the billing embeddings form cohesive, separated groups?*

Algorithm comparison

Five candidate methods are compared on four axes:

Criterion	What it measures
Silhouette	Cohesion within clusters, separation between clusters
Coverage	No large share of calls left unassigned (HDBSCAN noise handled)

Criterion	What it measures
Month stability	Silhouette stays consistent across all 7 months
Seed interpretability	How naturally cluster centroids align with known billing concepts

Winner: KMeans — full coverage, best silhouette, most stable across all months.

Silhouette sweep

Candidate cluster counts are evaluated over a range, with silhouette measured on a manageable embedding subsample. The initial notebook settled on **40 fine clusters** as the discovery resolution — not a permanent subcategory count.

> 40 was a useful window into the observed semantic space. It does not mean Brightspeed can only have 40 billing subcategories.

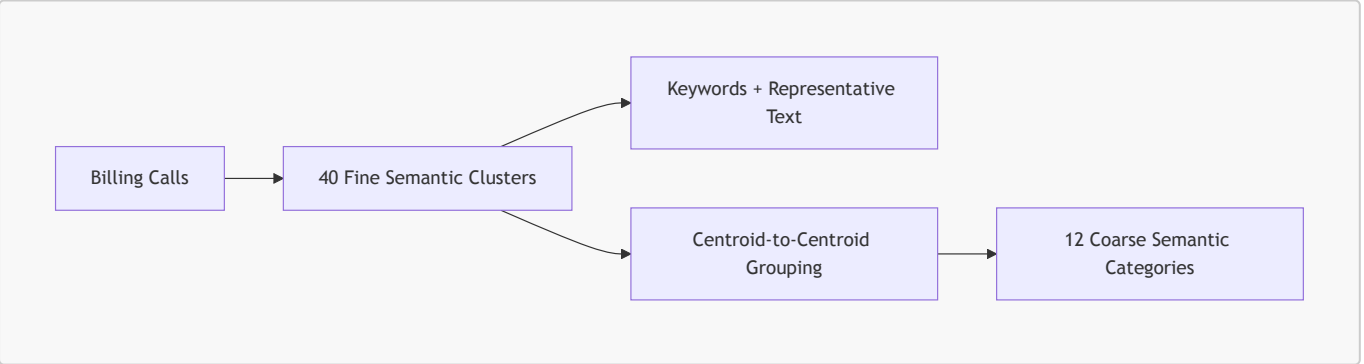
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4. Why 12 categories and 40 subcategories came first

Once 40 fine clusters existed, each was inspected for:

- Cluster volume
- Top TF-IDF keywords
- Representative text nearest centroid
- Semantic centroid vector

The 40 cluster centroids were then grouped into **12 coarse categories** using agglomerative clustering:



This was the first data-driven map of the billing topic space. It revealed major semantic neighborhoods, dominant volume drivers, confusing overlaps, and duplicate-looking clusters — but it did not yet define the permanent business vocabulary.

---

5. Scale-up: 80 fine clusters / 14 coarse groups

A later engineering build doubled the discovery resolution:

```
K_CLUSTERS = 80    K_CATEGORIES = 14    MIN_LEAF = 250
```

The purpose was a **stress test**: expose splits, overlaps, and fragmentation that a coarser model hides. Clusters below the minimum leaf size were folded rather than promoted. This improved discovery stability but created a production risk — genuinely rare but valid customer reasons could disappear.

---

6. The governance step — data-informed, not data-imprisoned

This is where the taxonomy was redesigned. A production ontology has three distinct concepts:

Status	Meaning
Discovered	Recurring evidence in the data forms a meaningful semantic topic
Observed Rare	Present in the data but too sparse for a robust unsupervised cluster
Coverage Expansion	A coherent Brightspeed billing reason with little or no observed evidence but that must remain classifiable
Fallback	Billing-related but no controlled leaf fits

Example: **Vacation hold billing** appears at volume 2 in aggregate evidence. Pure clustering would remove it. The production taxonomy retains it because it is linguistically distinct and operationally plausible.

---

7. The final taxonomy: 10 + 1 / 104

The production vocabulary was deliberately reshaped around **few macro categories and many specific customer asks**.

Category structure

#	Category	Leaves
1	Bill Understanding and Balance	8
2	Rates Discounts and Recurring Charges	8
3	Billing Errors Fees and Taxes	11
4	Payments and Payment Posting	12
5	AutoPay and Payment Methods	11
6	Payment Assistance Past Due and Collections	12
7	Credits Refunds and Adjustments	9
8	Billing Delivery Account and Preferences	9
9	Promotions Rewards and Gift Cards	6
10	Service Changes Equipment and Final Billing	17
11	Other Category (fallback)	1

#	Category	Leaves
	Total	**104**

Why the subcategory count grew from 40 → 104

The 104 leaves combine:

```
stable discovered intents
+ observed rare intents
+ deduplicated source reasons
+ explicit boundary splits
+ Brightspeed billing coverage expansions
+ one fallback
= controlled production vocabulary
```

Distinctions that deserve separate leaves even when clustering might merge them:

Cluster might merge…	Production separates…
Bill clarification	Bill dispute
Late-fee dispute	Late-fee waiver
Payment failure	Payment made but not posted
AutoPay failure	Generic payment failure
Final bill inquiry	Continued billing after cancellation
Past-due balance	Service restoration after payment
Vacation hold request	Vacation hold billing impact

The 104 leaves represent **business-semantic resolution**, not a claim that silhouette found 104 natural clusters.

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8. Canonical tags

Each production leaf has a stable machine-facing tag:

```
PAY_PAYMENT_NOT_POSTED
AUTO_AUTOPAY_PAYMENT_FAILURE
SVC_VACATION_HOLD_BILLING
ERR_INCORRECT_UNEXPECTED_CHARGE
OTHER_MISC_OTHER_BILLING
```

The canonical tag is the primary contract. Human-readable labels are obtained by lookup. The final taxonomy contains **104 canonical tags across 11 reporting categories**.

The mapped aggregate evidence file contains **111,836 source-reason observations** assigned to the final leaves. Coverage-only leaves can have zero historical evidence while remaining valid classifier outputs.

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9. GLVQ — prototype learning after discovery

Generalized Learning Vector Quantization learns a discriminative prototype for each active subcategory over the normalized embedding space.

Key distinction from KMeans:

	KMeans	GLVQ
Labels	None (unsupervised)	Cluster-to-taxonomy mapping
Goal	Discover geometric groups	Maximize inter-class separation
Output	Cluster assignments	Ranked prototypes with confidence

Training: 90/10 split · prototypes initialized from seed embeddings · gradient descent with logistic phi function · renormalized to unit sphere after every step.

Held-out accuracy measures **pseudo-label reproducibility**, not human ground-truth accuracy. Production validation should additionally measure:

- Manually reviewed top-1 exact leaf agreement
- Top-k recall on multi-topic examples
- Per-leaf precision and recall
- Fallback rate
- Month-over-month drift

GLVQ top-k softmax values are ranking scores, not calibrated correctness probabilities.

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10. SOM — semantic geography

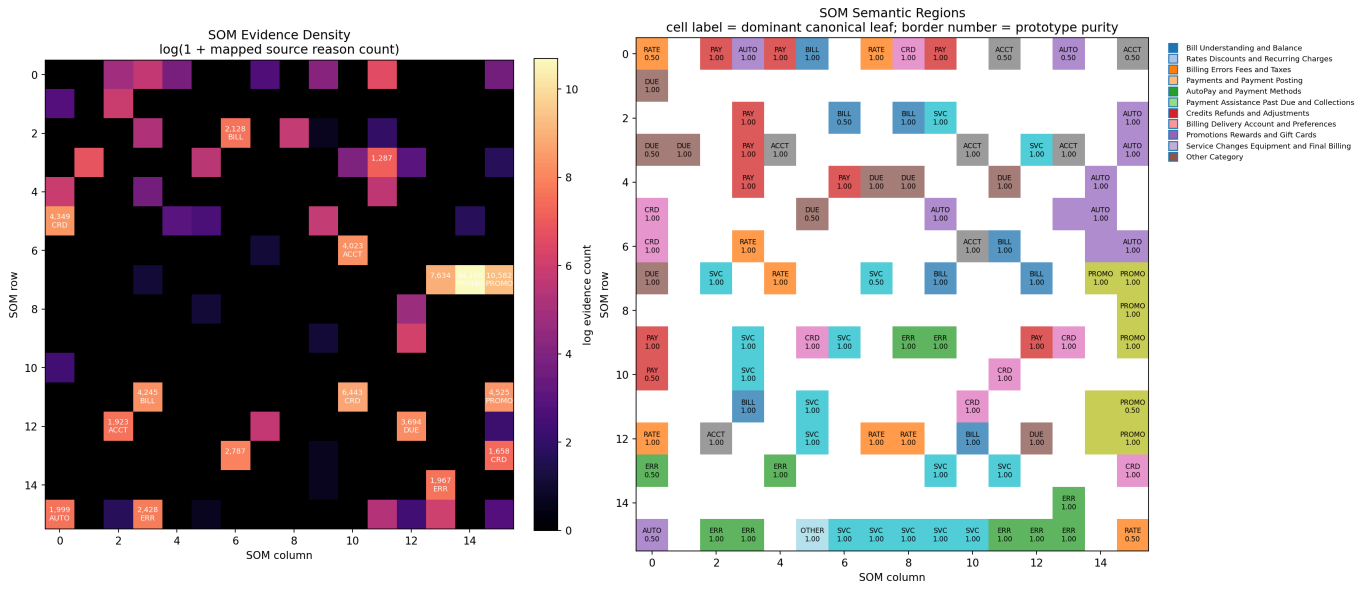
The Self-Organizing Map projects billing embeddings onto a **20×20 Kohonen grid**. Each call lands in its Best-Matching Unit (BMU) cell; cells accumulate counts to form a topographic density surface.

What a SOM answers that a taxonomy tree cannot:

- Which billing topics occupy nearby semantic regions?
- Where are dense concentrations of calls?
- Which categories have fuzzy borders?
- Which leaves are isolated and distinctive?
- Where might two labels be redundant?

SOM topology: evidence-density projection

Semantic prototype topology with mapped aggregate evidence projected onto cells. Cells are sized by evidence count and colored by dominant category.



SOM cell analytics (generated at runtime)

The notebook produces a second SOM view: a **20×20 heatmap** where cell color = BMU-aggregated call count (magma colormap) and white text shows the top TF-IDF keywords and call count per occupied cell. A companion **3D topographic surface** renders the same grid with height proportional to call density. Both outputs are generated by the pipeline (`som_cells.png`, `som_3d.png`) and can be inspected in `outputs/` after a full run.

Each cell carries:

```
cell(row, col) → call count · top keywords · dominant subcategory tag
```

For example (from the mapped evidence):

Cell	Dominant tag	Category	Evidence count
(0, 3)	AUTO_CONFIRM_AUTOPAY_ENROLLMENT	AutoPay and Payment Methods	296
(0, 11)	ACCT_PAPERLESS_BILLING_GENERAL_INQUIRY	Billing Delivery Account and Preferences	678
(2, 6)	BILL_PAYMENT_DUE_DATE	Bill Understanding and Balance	2,128
(2, 8)	BILL_CURRENT_ACCOUNT_BALANCE	Bill Understanding and Balance	318
(3, 1)	DUE_COLLECTIONS_NOTICE_AGENCY_INQUIRY	Payment Assistance Past Due and Collections	941

> **Note:** The bundled SOM evidence-density image is a **semantic prototype topology with mapped aggregate evidence**, not a raw 832,885-call BMU density map. The original notebook's SOM update implementation differs from the conventional neighborhood update; the map is therefore a diagnostic visualization.

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11. Treemap — evidence density in the production hierarchy

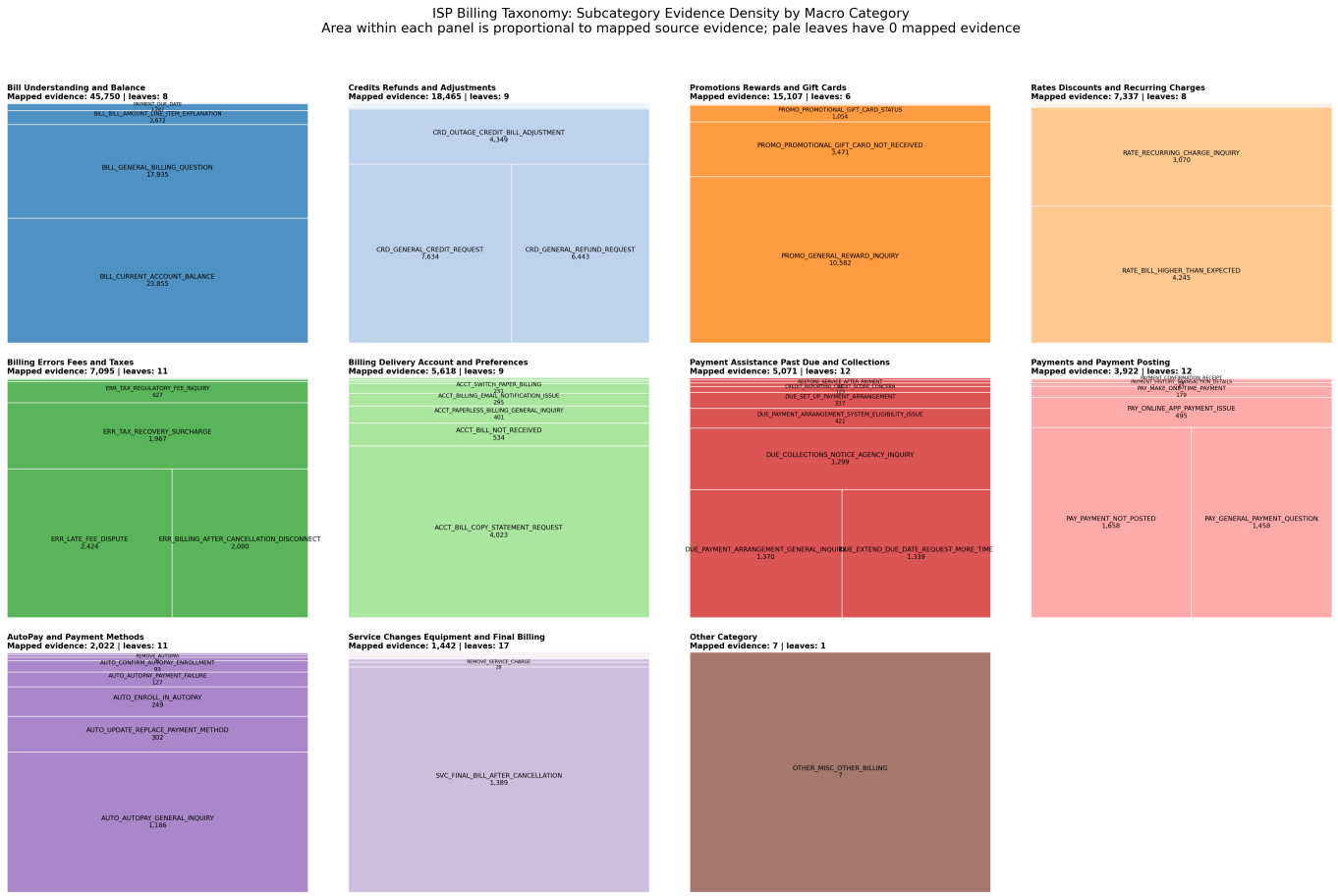
Subcategory call-volume treemap

The full subcategory treemap shows every leaf sized by its call volume, colored by macro category. Each tile carries the leaf name and raw count. This is the canonical density view of the 104-leaf taxonomy.



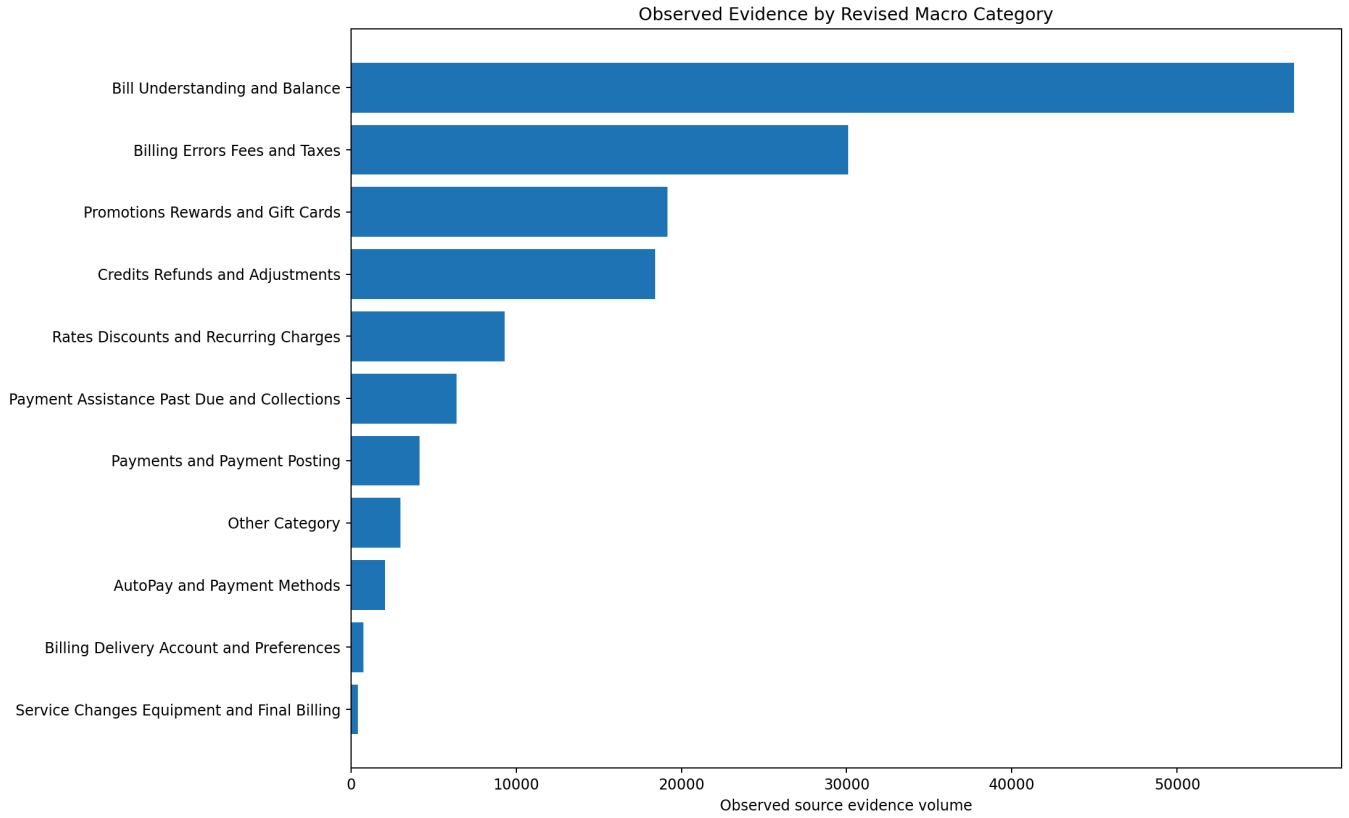
Subcategory evidence-density treemap

A second treemap overlays mapped aggregate source-reason evidence onto the same hierarchy. Observed high-volume reasons occupy more area. Low-volume or coverage-only leaves remain present even when their observed area is very small.



Evidence by category

Macro-level rollup showing which categories absorb the most observed billing evidence.



This prevents two common errors:

1. **Removing valid rare reasons** because their current volume is low
2. **Visually inflating coverage-only labels** and pretending they were discovered at high volume

---

12. Inference contract

The production classifier does not force every call to have three topics.

```
tag_1  =  mandatory
tag_2  =  optional (separate additional billing ask only)
tag_3  =  optional (separate additional billing ask only)
```

Do not create extra tags for:

- Synonyms or restatements
- Causes or consequences of the same problem
- Background account facts
- Alternate plausible interpretations

This is an entropy-reduction choice. Rank order communicates useful information without requiring pseudo-probability scores. The fallback `OTHER_MISC_OTHER_BILLING` is a valid primary tag when no specific leaf fits.

---

13. Population-level tagging results

Metric	Count	Share
Billing rows tagged	**832,885**	100.0%
Exactly 1 topic	**784,777**	94.2%
Exactly 2 topics	**45,288**	5.4%
3 topics	**2,820**	0.3%
At least a second topic	**48,108**	5.8%
Using `OTHER_MISC_OTHER_BILLING`	**71,922**	8.6%

Low rank-2 and rank-3 rates are intentional — the system recognizes genuine multi-topic contacts without turning uncertainty into extra labels.

Controlled-taxonomy summary

Metric	Value
Macro categories (including fallback)	11
Canonical subcategories	104
Curated centroid texts per leaf	10

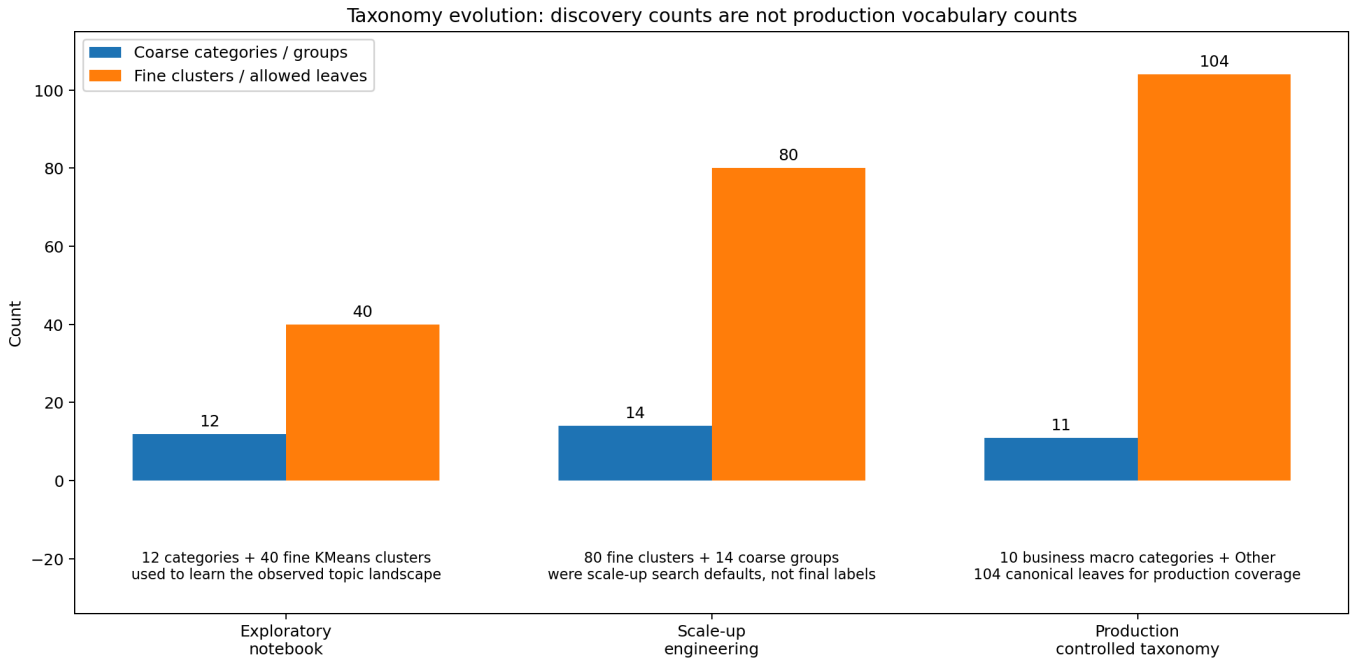
Metric	Value
Total curated centroid texts	1,040
Mapped aggregate source evidence	111,836
Final YTD rows scored	832,885

14. Subcategories per macro category

Category	Leaf count
Bill Understanding and Balance	8
Rates Discounts and Recurring Charges	8
Billing Errors Fees and Taxes	11
Payments and Payment Posting	12
AutoPay and Payment Methods	11
Payment Assistance Past Due and Collections	12
Credits Refunds and Adjustments	9
Billing Delivery Account and Preferences	9
Promotions Rewards and Gift Cards	6
Service Changes Equipment and Final Billing	17
Other Category	1
Total	**104**

15. Complete taxonomy tree

The full 104-leaf hierarchy with canonical tags, definitions, and evidence provenance:



Bill Understanding and Balance (8 leaves)

Tag	Subcategory	Evidence
BILL_GENERAL_BILLING_QUESTION	General billing question	Observed
BILL_CURRENT_ACCOUNT_BALANCE	Current account balance	Observed
BILL_PAYMENT_DUE_DATE	Payment due date	Observed
BILL_BILL_AMOUNT_LINE_ITEM_EXPLANATION	Bill amount or line item explanation	Observed
BILL_FIRST_BILL_EXPLANATION	First bill explanation	Coverage Expansion
BILL_PRORATED_BILL_EXPLANATION	Prorated bill explanation	Coverage Expansion
BILL_BILLING_CYCLE_STATEMENT_DATE	Billing cycle or statement date	Coverage Expansion
BILL_UPCOMING_ESTIMATED_BILL_AMOUNT	Upcoming or estimated bill amount	Coverage Expansion

Rates Discounts and Recurring Charges (8 leaves)

Tag	Subcategory	Evidence
RATE_BILL_HIGHER_THAN_EXPECTED	Bill higher than expected	Observed
RATE_PRICE_RATE_INCREASE	Price or rate increase	Observed
RATE_PROMOTIONAL_RATE_EXPIRED	Promotional rate expired	Coverage Expansion

Tag	Subcategory	Evidence
RATE_MISSING_INCORRECT_DISCOUNT	Missing or incorrect discount	Coverage Expansion
RATE_LOWER_PRICE_DISCOUNT_REQUEST	Lower price or discount request	Observed
RATE_PLAN_PACKAGE_PRICE_INQUIRY	Plan or package price inquiry	Observed
RATE_RECURRING_CHARGE_INQUIRY	Recurring charge inquiry	Observed
RATE_ADD_FEATURE_RECURRING_CHARGE	Add-on or feature recurring charge	Coverage Expansion

Billing Errors Fees and Taxes (11 leaves)

Tag	Subcategory	Evidence
ERR_INCORRECT_UNEXPECTED_CHARGE	Incorrect or unexpected charge	Observed
ERR_DUPLICATE_CHARGE	Duplicate charge	Coverage Expansion
ERR_UNAUTHORIZED_BILLING_CHARGE	Unauthorized billing charge	Coverage Expansion
ERR_BILLING_AFTER_CANCELLATION_DISCONNECT	Billing after cancellation or disconnect	Observed
ERR_LATE_FEE_DISPUTE	Late fee dispute	Observed
ERR_LATE_FEE_WAIVER_AVOIDANCE	Late fee waiver or avoidance	Observed
ERR_RETURNED_PAYMENT_INSUFFICIENT_FUNDS_FEE	Returned payment or insufficient funds fee	Coverage Expansion
ERR_RECONNECTION_RESTORATION_FEE	Reconnection or restoration fee	Coverage Expansion
ERR_TAX_REGULATORY_FEE_INQUIRY	Tax or regulatory fee inquiry	Observed
ERR_TAX_RECOVERY_SURCHARGE	Tax recovery surcharge	Observed
ERR_INCORRECT_TAX_FEE_AMOUNT	Incorrect tax or fee amount	Coverage Expansion

Payments and Payment Posting (12 leaves)

Tag	Subcategory	Evidence
PAY_MAKE_ONE_TIME_PAYMENT	Make a one-time payment	Observed

Tag	Subcategory	Evidence
PAY_GENERAL_PAYMENT_QUESTION	General payment question	Observed
PAY_ONLINE_APP_PAYMENT_ISSUE	Online or app payment issue	Observed
PAY_PAYMENT_DECLINED_FAILED	Payment declined or failed	Coverage Expansion
PAY_PAYMENT_NOT_POSTED	Payment not posted	Observed
PAY_PAYMENT_APPLIED_WRONG_ACCOUNT	Payment applied to wrong account	Observed
PAY_PAYMENT_CONFIRMATION_RECEIPT	Payment confirmation or receipt	Observed
PAY_PAYMENT_REVERSED_RETURNED	Payment reversed or returned	Coverage Expansion
PAY_DUPLICATE_PAYMENT	Duplicate payment	Coverage Expansion
PAY_PARTIAL_PAYMENT	Partial payment	Coverage Expansion
PAY_CHECK_MAILED_PAYMENT_STATUS	Check or mailed payment status	Observed
PAY_PAYMENT_HISTORY_TRANSACTION_DETAILS	Payment history or transaction details	Observed

AutoPay and Payment Methods (11 leaves)

Tag	Subcategory	Evidence
AUTO_AUTOPAY_GENERAL_INQUIRY	AutoPay general inquiry	Observed
AUTO_ENROLL_IN_AUTOPAY	Enroll in AutoPay	Observed
AUTO_CONFIRM_AUTOPAY_ENROLLMENT	Confirm AutoPay enrollment	Observed
AUTO_REMOVE_AUTOPAY	Remove AutoPay	Observed
AUTO_AUTOPAY_PAYMENT_FAILURE	AutoPay payment failure	Observed
AUTO_AUTOPAY_CHARGED_WRONG_AMOUNT	AutoPay charged wrong amount	Coverage Expansion
AUTO_DUPLICATE_AUTOPAY_WITHDRAWAL	Duplicate AutoPay withdrawal	Coverage Expansion
AUTO_CHANGE_AUTOPAY_PAYMENT_DATE	Change AutoPay payment date	Coverage Expansion

Tag	Subcategory	Evidence
AUTO_UPDATE_REPLACE_PAYMENT_METHOD	Update or replace payment method	Observed
AUTO_UPDATE_AUTOPAY_PAYMENT_METHOD	Update AutoPay payment method	Observed
AUTO_COMPROMISED_CARD_BANK_ACCOUNT	Compromised card or bank account	Observed

Payment Assistance Past Due and Collections (12 leaves)

Tag	Subcategory	Evidence
DUE_PAYMENT_ARRANGEMENT_GENERAL_INQUIRY	Payment arrangement general inquiry	Observed
DUE_SET_UP_PAYMENT_ARRANGEMENT	Set up payment arrangement	Observed
DUE_EXTEND_DUE_DATE_REQUEST_MORE_TIME	Extend due date or request more time	Observed
DUE_PAYMENT_ARRANGEMENT_SYSTEM_ELIGIBILITY_ISSUE	Payment arrangement system or eligibility issue	Observed
DUE_MISSED_BROKEN_PAYMENT_ARRANGEMENT	Missed or broken payment arrangement	Coverage Expansion
DUE_FINANCIAL_HARDSHIP_PAYMENT_ASSISTANCE	Financial hardship or payment assistance	Coverage Expansion
DUE_PAST_DUE_BALANCE_NONPAYMENT	Past due balance or nonpayment	Observed
DUE_SERVICE_SUSPENDED_DISCONNECTED_NONPAYMENT	Service suspended for nonpayment	Observed
DUE_RESTORE_SERVICE_AFTER_PAYMENT	Restore service after payment	Observed
DUE_COLLECTIONS_NOTICE_AGENCY_INQUIRY	Collections notice or agency inquiry	Observed
DUE_CREDIT_REPORTING_CREDIT_SCORE_CONCERN	Credit reporting or credit score concern	Observed
DUE_PAID_DEBT_STILL_IN_COLLECTIONS	Paid debt still in collections	Coverage Expansion

Credits Refunds and Adjustments (9 leaves)

Tag	Subcategory	Evidence
CRD_GENERAL_CREDIT_REQUEST	General credit request	Observed
CRD_GENERAL_REFUND_REQUEST	General refund request	Observed
CRD_OUTAGE_CREDIT_BILL_ADJUSTMENT	Outage credit or bill adjustment	Observed
CRD_SERVICE_QUALITY_CREDIT	Service quality credit	Coverage Expansion
CRD_MISSING_CREDIT_ADJUSTMENT	Missing credit or adjustment	Coverage Expansion
CRD_REFUND_NOT_RECEIVED_REFUND_STATUS	Refund not received or refund status	Observed
CRD_REFUND_METHOD_TIMING_INQUIRY	Refund method or timing inquiry	Coverage Expansion
CRD_OVERPAYMENT_REFUND	Overpayment refund	Coverage Expansion
CRD_DEPOSIT_REFUND	Deposit refund	Coverage Expansion

Billing Delivery Account and Preferences (9 leaves)

Tag	Subcategory	Evidence
ACCT_BILL_COPY_STATEMENT_REQUEST	Bill copy or statement request	Observed
ACCT_BILL_NOT_RECEIVED	Bill not received	Coverage Expansion
ACCT_PAPERLESS_BILLING_GENERAL_INQUIRY	Paperless billing general inquiry	Observed
ACCT_ENROLL_IN_PAPERLESS_BILLING	Enroll in paperless billing	Observed
ACCT_SWITCH_PAPER_BILLING	Switch to paper billing	Observed
ACCT_BILLING_EMAIL_NOTIFICATION_ISSUE	Billing email or notification issue	Coverage Expansion
ACCT_ONLINE_ACCOUNT_ACCESS_VERIFICATION	Online account access or verification	Observed
ACCT_UPDATE_BILLING_ADDRESS	Update billing address	Observed
ACCT_UPDATE_BILLING_CONTACT_INFORMATION	Update billing contact information	Coverage Expansion

Promotions Rewards and Gift Cards (6 leaves)

Tag	Subcategory	Evidence
PROMO_GENERAL_REWARD_INQUIRY	General reward inquiry	Observed
PROMO_PROMOTIONAL_GIFT_CARD_NOT_RECEIVED	Promotional gift card not received	Observed
PROMO_PROMOTIONAL_GIFT_CARD_STATUS	Promotional gift card status	Observed
PROMO_GIFT_CARD_AMOUNT_VALUE_INQUIRY	Gift card amount or value inquiry	Coverage Expansion
PROMO_PROMOTION_REWARD_ELIGIBILITY	Promotion or reward eligibility	Coverage Expansion
PROMO_PROMOTION_TERMS_FULFILLMENT_TIMING	Promotion terms or fulfillment timing	Coverage Expansion

Service Changes Equipment and Final Billing (17 leaves)

Tag	Subcategory	Evidence
SVC_FINAL_BILL_AFTER_CANCELLATION	Final bill after cancellation	Observed
SVC_CANCEL_DISCONNECT_SERVICE_WITH_BILLING_IMPACT	Cancel or disconnect with billing impact	Observed
SVC_REMOVE_SERVICE_CHARGE	Remove service or charge	Observed
SVC_VACATION_HOLD_BILLING	Vacation hold billing	Observed Rare
SVC_RESUME_SERVICE_AFTER_VACATION_HOLD_BILLING	Resume service after vacation hold	Coverage Expansion
SVC_MOVE_TRANSFER_SERVICE_BILLING	Move or transfer service billing	Coverage Expansion
SVC_UPGRADE_DOWNGRADE_BILLING_IMPACT	Upgrade or downgrade billing impact	Observed
SVC_ADD_SERVICE_FEATURE_BILLING_IMPACT	Add service or feature billing impact	Coverage Expansion
SVC_CHANGE_EFFECTIVE_DATE_PRORATION	Change effective date or proration	Coverage Expansion

Tag	Subcategory	Evidence
SVC_CLOSED_ACCOUNT_BALANCE_FINAL_PAYMENT	Closed account balance or final payment	Coverage Expansion
SVC_MODEM_ROUTER_RENTAL_FEE	Modem or router rental fee	Observed
SVC_EQUIPMENT_PURCHASE_OWNERSHIP_CHARGE	Equipment purchase or ownership charge	Observed
SVC_EQUIPMENT_RETURN_NONRETURN_CHARGE	Equipment return or nonreturn charge	Coverage Expansion
SVC_INSTALLATION_ACTIVATION_CHARGE	Installation or activation charge	Coverage Expansion
SVC_TECHNICIAN_VISIT_SERVICE_CALL_CHARGE	Technician visit or service call charge	Coverage Expansion
SVC_EQUIPMENT_SHIPPING_HANDLING_CHARGE	Equipment shipping or handling charge	Coverage Expansion
SVC_REPLACEMENT_EQUIPMENT_CHARGE	Replacement equipment charge	Coverage Expansion

Other Category (1 leaf)

Tag	Subcategory	Evidence
OTHER_MISC_OTHER_BILLING	Misc or Other billing	Fallback

16. Excel inference workflow

The taxonomy is deployed to stakeholders through an Excel-based workflow. The key artifacts:

Artifact	File	Purpose
Taxonomy workbook	billing_taxonomy_workbook.xlsx	Review workbook containing the canonical vocabulary, prompt, and non-technical usage notes
Inference prompt	billing_inference_prompt.txt	Compact classification prompt for LLM-based or manual tagging

Artifact	File	Purpose
ChatGPT Plugin prompt	`ChatGPT_Prompt.md`	Full controlled-vocabulary prompt for business users self-serving via the ChatGPT Excel plugin. Contains anti-drift rules, the complete 104-tag vocabulary table, and usage instructions
Example output	`brightspeed_excel_inference_output.xlsx`	Worked inference output showing tag assignment on sample rows
Canonical vocabulary	`canonical_vocabulary.csv`	Allowed tag vocabulary with exact readable labels

Classification prompt

The prompt used for Excel-based classification (`billing_inference_prompt.txt`):

```
BRIGHTSPEED BILLING CALL CLASSIFICATION

Classify each call using only the call_intent text and the
Canonical_Vocabulary sheet.

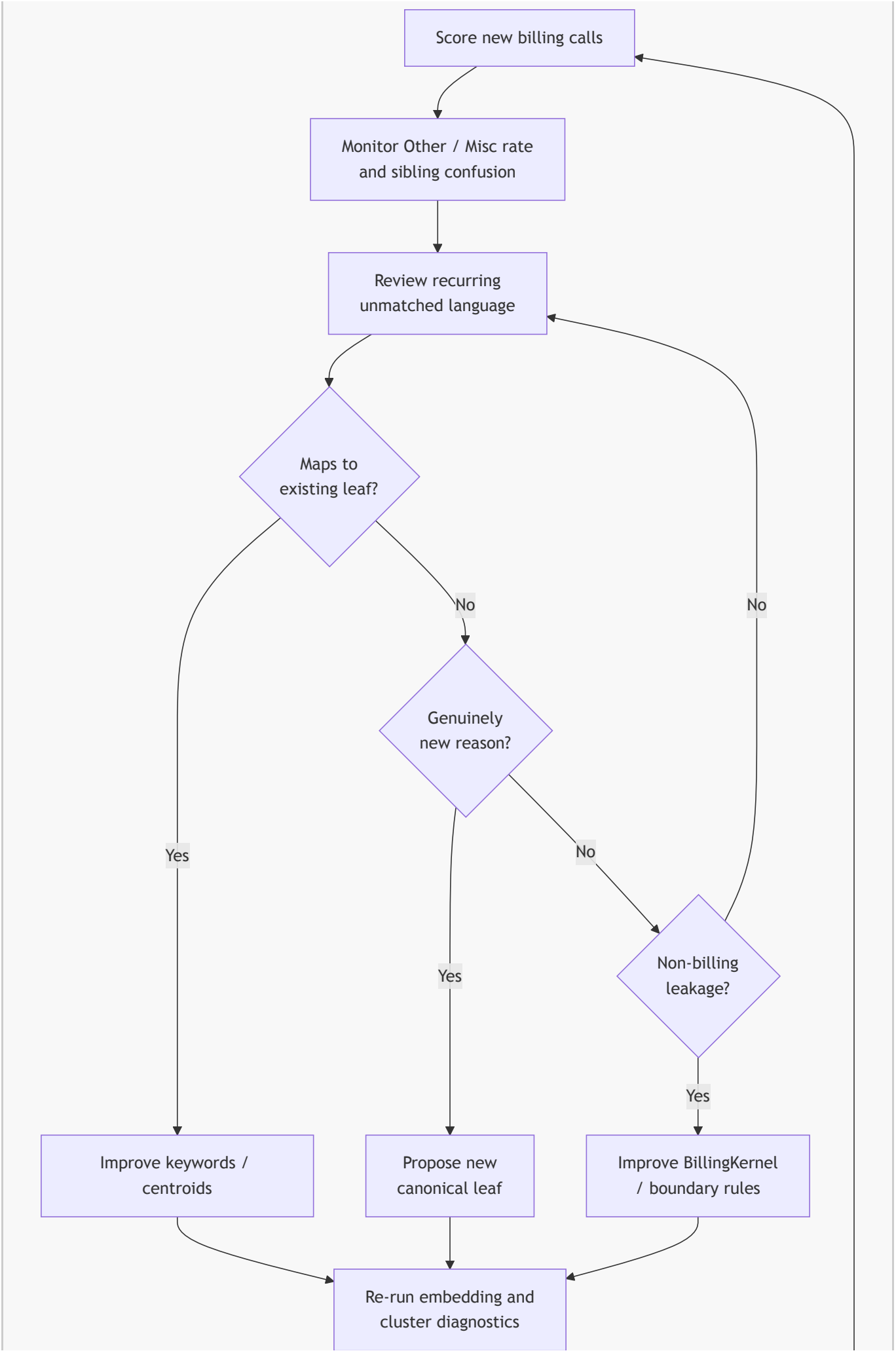
RULES
1. tag_1 is required. Choose the single most specific supported
canonical_tag.
2. Use tag_2 or tag_3 only when the call clearly contains a separate
   additional billing ask. Leave unused ranks blank.
3. Do not add extra tags for synonyms, background facts, causes,
   symptoms, or consequences of the same issue.
4. Rank distinct topics by the customer's main ask, then secondary asks.
5. Use only canonical_tag values from Canonical_Vocabulary.
   Never invent or alter a tag.
6. If no specific tag fits, use OTHER_MISC_OTHER_BILLING.
7. Copy category and subcategory exactly from Canonical_Vocabulary
   for each selected tag.

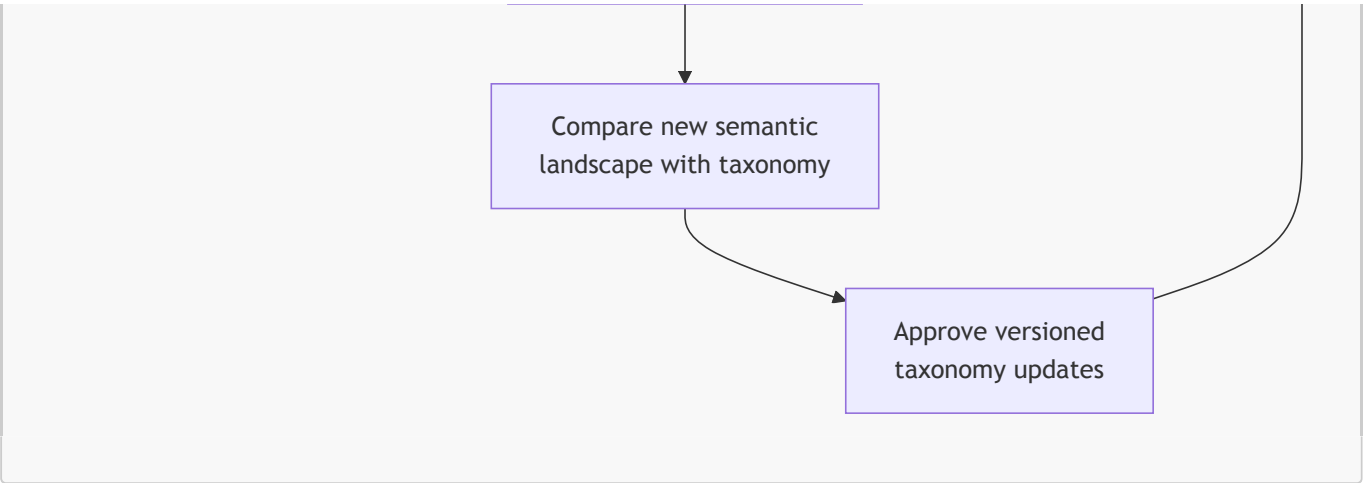
WRITE THESE COLUMNS
ai_tag_1, ai_category_1, ai_subcategory_1
ai_tag_2, ai_category_2, ai_subcategory_2
ai_tag_3, ai_category_3, ai_subcategory_3

Leave rank 2 and rank 3 fields blank unless the call truly has
multiple distinct billing topics.
```

---

18. Recommended governance loop


Score new billing calls
Monitor Other / Misc rate and sibling confusion
Review recurring unmatched language
Maps to existing leaf?
Yes
No
Genuinely new reason?
Yes
No
Non-billing leakage?
Yes
No
Improve keywords / centroids
Propose new canonical leaf
Improve BillingKernel / boundary rules
Re-run embedding and cluster diagnostics



High-volume structural changes should be supported by data. Rare but legitimate customer reasons should not disappear because a clustering threshold cannot discover them in the current period.

---

19. What the process proves — and what it does not

Supported conclusions

Brightspeed billing contacts contain a stable core of recurring semantic themes around bill understanding, pricing, errors, payments, AutoPay, payment assistance, credits, billing preferences, promotions, and service-lifecycle billing consequences.

The discovery workflow provides empirical evidence for those neighborhoods. The final taxonomy expands that evidence into a governed production vocabulary.

Honest limitations

Claim	Reality
104 silhouette-validated natural clusters	**No.** 104 business-semantic leaves — not all silhouette-optimal
10 mathematically optimal macro categories	**No.** 10 reporting categories chosen for operational usefulness
GLVQ accuracy = ground truth	**No.** Held-out accuracy measures pseudo-label reproducibility only
GLVQ scores = calibrated probabilities	**No.** Top-k scores are ranking scores, not correctness probabilities
SOM = raw 832K-call BMU map	**No.** Evidence-density prototype topology with mapped aggregate evidence

These distinctions are strengths, not weaknesses. They keep discovery, modeling, governance, and production classification appropriately separated.

Bottom line

The **12-category / 40-cluster** result was the right starting point — it revealed the semantic skeleton. The **10 macro categories + Other / 104 canonical leaves** is the right production output — it adds governance, billing boundaries, rare-intent retention, canonical machine tags, and a fallback.

The notebook, algorithm search, silhouette analysis, GLVQ, SOM, treemap, controlled taxonomy, and YTD inference are not competing approaches. They are sequential layers of one process:

> **Discover the landscape → test the geometry → govern the vocabulary → validate the boundaries → visualize the evidence → classify at scale.**

Appendix — artifact inventory

All production artifacts live in `artifacts/`. Visual assets for this report live in `assets/`.

Taxonomy and vocabulary

File	Description
`taxonomy.json`	Full seeded taxonomy hierarchy (categories → subcategories → keywords)
`canonical_vocabulary.csv`	Allowed tag vocabulary: canonical tag, category, subcategory, definition, keywords
`tree_hierarchy.md`	Complete 104-leaf taxonomy tree with definitions and evidence provenance
`taxonomy_keywords.csv`	Per-leaf TF-IDF keyword sets
`curated_centroids_10_each.csv`	10 curated representative texts per leaf (1,040 total)

Evidence and mapping

File	Description
`source_reason_to_canonical_mapping.csv`	Maps every source reason text to its canonical leaf tag
`canonical_leaf_evidence_counts.csv`	Observed evidence count per canonical leaf (111,836 mapped rows)
`som_cells_evidence_density.csv`	SOM cell-level data: cell coords, dominant tag, evidence count, purity
`som_legend.csv`	SOM cell → subcategory mapping (20×20 grid legend)

Visualizations

File	Description
treemap.png	Call-volume treemap by Category → Subcategory
treemap_subcategories.png	Subcategory treemap with leaf-level call counts
treemap_subcategory_evidence_density.png	Evidence-density treemap overlay
evidence_by_category.png	Macro-category evidence rollup
som_evidence_density.png	SOM evidence-density topology (report visual)
som_map.png	SOM 20×20 cell-level map with keywords

Excel workflow

File	Description
billing_taxonomy_workbook.xlsx	Review workbook with taxonomy, prompt, and usage notes
billing_inference_prompt.txt	Compact classification prompt for Excel-based tagging
ChatGPT_Prompt.md	Full controlled-vocabulary prompt for ChatGPT Excel plugin with anti-drift rules
brightspeed_excel_inference_output.xlsx	Worked inference output example

Model artifacts (generated by pipeline)

File	Description
billing_kernel.joblib	Trained BillingKernel binary classifier
glvq.joblib	Trained GLVQ prototype model
glvq_active_keys.json	Active subcategory keys for GLVQ scoring
som20.joblib / som.joblib	Trained 20×20 Kohonen SOM
subcat_seed_protos.npy	Embedded subcategory seed prototypes
subcat_artifacts.json	Subcategory artifact metadata